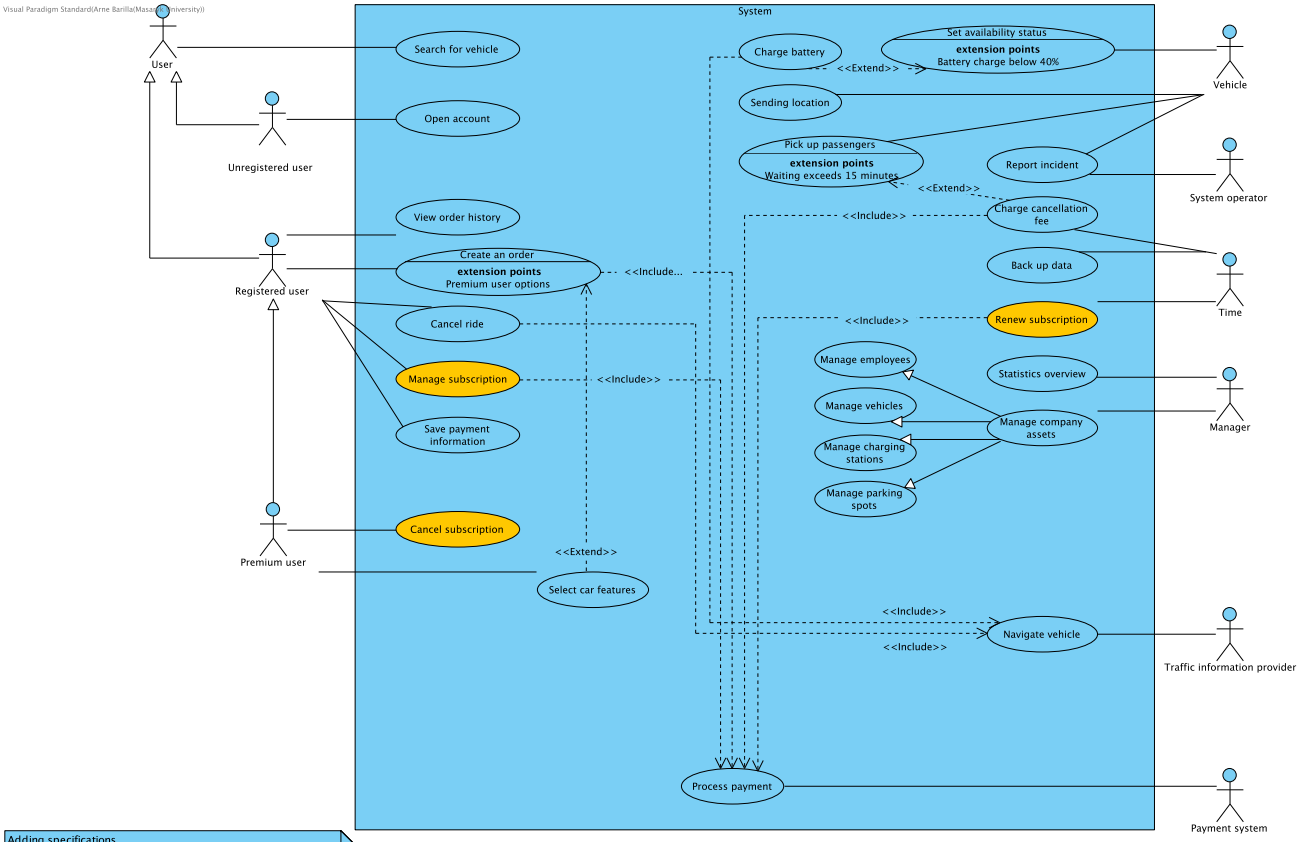
1. SmartTaxIS – Use Case Diagram



Functional Requirements

User
FR1. The system allows searching for a vehicle.

Unregistered User
FR2. The system allows the creation of an account.

Registered User
FR3. The system allows the creation of an order.
FR4. The system allows browsing the order history.
FR5. The system allows users to save their payment details.
FR6. The system allows the purchase of a premium membership.
FR7. The system allows a passenger to cancel the ride.

Premium User
FR8. The system allows the selection of vehicle equipment.
FR9. The system allows cancelling the subscription.

Vehicle
FR10. The system tracks the current location of the vehicle.

FR11. The system monitors the availability status of the vehicle.
FR12. The system computes routes and navigates the vehicle.
FR13. The system notifies the user when the vehicle arrives at the pickup location.

Manager
FR14. The system allows the manager to manage the company's assets.
FR15. The system displays statistics.

System operator
FR16. The system notifies the system operator about incidents during a route.

Time
FR17. The system charges a fee for order cancellation.
FR18. The system backs up data every day at 23:59.
FR19. The system automatically renews subscriptions.

Payment system
FR20. The external payment system executes payments.

Non-Functional Requirements
NFR1. The vehicles managed by the system are electric and fully autonomous.
NFR2. Each vehicle sends its position to the system at least once per second.
NFR3. The system must be available 24/7.
NFR4. The system must display vehicle search results within three seconds.
NFR5. Stored payment card information must be encrypted and handled securely.

1.1. Back up data

ID: UC28

Use Case ID	UC28
Brief Description	System backs up all data at 23:59 everyday.
Primary Actors	
Secondary Actors	
Preconditions	
Main Flow of Events	
Alternative Flows	
Post-conditions	

1.2. Cancel ride

ID: UC34

Use Case ID	UC34

Brief Description	A passenger can request cancellation of the ride at any time. The system finds a suitable place to park, updates the vehicle's destination and sends the new route to the vehicle.
Primary Actors	
Secondary Actors	
Preconditions	
Main Flow of Events	
Alternative Flows	
Post-conditions	

1.3. Cancel subscription

ID: UC26

Use Case ID	UC26
Brief Description	A premium user can cancel the subscription at any time. The premium benefits remain active until the end of the current billing period, after which the subscription is not renewed and the user becomes a registered user again.
Primary Actors	Premium user
Secondary Actors	-
Preconditions	1. The user is logged in to the system. 2. The user has an active premium subscription.
Main Flow of Events	1. The premium user selects the option to cancel the subscription. 2. The system displays the end date of the current billing period and informs the user that the premium benefits remain active until then. 3. The system asks the user to confirm the cancellation. 4. The premium user confirms the cancellation. 5. The system marks the subscription as cancelled so that it will not be renewed. 6. The system confirms the cancellation to the user.
Alternative Flows	4a. The user does not confirm the cancellation 1. The premium user leaves the cancellation dialog. 2. The subscription remains active and will be renewed as usual.
Post-conditions	1. The subscription is marked as cancelled. 2. The user keeps the premium benefits until the end of the current billing period, then loses premium status (Renew subscription, alternative flow 3a).

1.4. Charge battery

ID: UC12

Use Case ID	UC12
Brief Description	When vehicle's battery charge level drops below 40%, it is navigated to the nearest free charging station.
Primary Actors	
Secondary Actors	
Preconditions	
Main Flow of Events	
Alternative Flows	
Post-conditions	

1.5. Charge cancellation fee

ID: UC29

Use Case ID	UC29
Brief Description	When the vehicle arrives at the pickup location, it waits until all passengers get on board. If the waiting time exceeds 15 minutes, the system automatically cancels the order and charges the user a cancellation fee through the external payment system.
Primary Actors	
Secondary Actors	
Preconditions	
Main Flow of Events	
Alternative Flows	
Post-conditions	

1.6. Create an order

ID: UC02

Use Case ID	UC02
Brief Description	Registered user can create an order. Registered user confirms one of the offers found by the vehicle search. Unregistered users can search for vehicles but must create an account before confirming an order.
Primary Actors	
Secondary Actors	
Preconditions	
Main Flow of Events	
Alternative Flows	
Post-conditions	

1.7. Manage charging stations

ID: UC32

Use Case ID	UC32
Brief Description	The manager can create, update and delete charging stations managed by the system. For each station, the manager sets its location, the maximum number of vehicles that can charge at the same time and its maximum power output. This use case is a specialization of Manage company assets.
Primary Actors	
Secondary Actors	
Preconditions	
Main Flow of Events	
Alternative Flows	
Post-conditions	

1.8. Manage company assets

ID: UC14

Use Case ID	UC14

Brief Description	The manager can manage all assets associated with the taxi company, i.e. create, update and delete their records. This is a general use case specialized by Manage employees, Manage vehicles, Manage charging stations and Manage parking spots.
Primary Actors	
Secondary Actors	
Preconditions	
Main Flow of Events	
Alternative Flows	
Post-conditions	

1.9. Manage employees

ID: UC30

Use Case ID	UC30
Brief Description	The manager can create, update and delete employee records in the system. Each record holds the employee's personal and contact details and their role in the company (e.g. system operator). This use case is a specialization of Manage company assets.
Primary Actors	
Secondary Actors	
Preconditions	
Main Flow of Events	
Alternative Flows	
Post-conditions	

1.10. Manage parking spots

ID: UC33

Use Case ID	UC33

Brief Description	The manager can create, update and delete the parking spots reserved for SmartTaxIS vehicles. For each spot, the manager records its location, and the system then monitors its availability. This use case is a specialization of Manage company assets.
Primary Actors	
Secondary Actors	
Preconditions	
Main Flow of Events	
Alternative Flows	
Post-conditions	

1.11. Manage subscription

ID: UC08

Use Case ID	UC08
Brief Description	A registered user can subscribe to a premium membership and become a premium user. The user chooses whether the subscription is billed monthly or annually and pays the first period through the external payment system. Renewal of the subscription is handled by Renew subscription.
Primary Actors	Registered user
Secondary Actors	Payment system
Preconditions	1. The user is registered and logged in to the system. 2. The user does not have an active premium subscription.
Main Flow of Events	1. The registered user selects the option to purchase a premium membership. 2. The system displays the premium benefits (broader search options, 20 % discount on all orders) and the available billing options (monthly / annually) with their prices. 3. The registered user selects the billing option. 4. The system displays a summary of the subscription and asks the user to confirm it. 5. The registered user confirms the subscription. 6. INCLUDE: Process payment - the system charges the user for the first billing period using the stored payment card. 7. The system activates the premium membership and sets the end date of the current billing period. 8. The system informs the user that the subscription is active and will be renewed automatically.

Alternative Flows	3a. The user cancels the purchase 1. The user leaves the purchase before confirming it. 2. The use case ends without any changes. 6a. The user has no stored payment card 1. The system asks the user to enter payment card details. 2. The user enters the details, which are saved in the system. 3. The use case continues at step 6. 6b. The payment fails 1. The system informs the user that the payment was not successful. 2. The premium membership is not activated and the use case ends.
Post-conditions	1. The user is a premium user and can use the premium benefits. 2. The subscription will be renewed automatically (Renew subscription).

1.12. Manage vehicles

ID: UC31

Use Case ID	UC31
Brief Description	The manager can add new vehicles to the fleet, update their details and remove vehicles that are no longer in service. For each vehicle, the manager records its maximum passenger capacity and installed additional features (e.g. entertainment systems, comfort accessories). This use case is a specialization of Manage company assets.
Primary Actors	
Secondary Actors	
Preconditions	
Main Flow of Events	
Alternative Flows	
Post-conditions	

1.13. Manager

ID: AC06

1.14. Navigate vehicle

ID: UC35

Use Case ID	UC35

Brief Description	The system computes a route as a collection of waypoints based on the current position of the vehicle and the traffic situation obtained from the external traffic information provider. The route is sent to the vehicle, which follows it autonomously.
Primary Actors	
Secondary Actors	
Preconditions	
Main Flow of Events	
Alternative Flows	
Post-conditions	

1.15. Open account

ID: UC01

Use Case ID	UC01
Brief Description	Unregistered user can create a new account.
Primary Actors	
Secondary Actors	
Preconditions	
Main Flow of Events	
Alternative Flows	
Post-conditions	

1.16. Payment system

ID: AC08

1.17. Pick up passengers

ID: UC36

Use Case ID	UC36

Brief Description	When the vehicle arrives at the pickup location, the system notifies the user. The vehicle waits until all passengers get on board and then heads to the destination.
Primary Actors	
Secondary Actors	
Preconditions	
Main Flow of Events	
Alternative Flows	
Post-conditions	

1.18. Premium user

ID: AC04

1.19. Process payment

ID: UC27

Use Case ID	UC27
Brief Description	This use case allows the system to charge the user for a service through an external online payment provider.
Primary Actors	
Secondary Actors	
Preconditions	
Main Flow of Events	
Alternative Flows	
Post-conditions	

1.20. Registered user

ID: AC03

1.21. Renew subscription

ID: UC37

Use Case ID	UC37

Brief Description	Three days before the end of the current billing period, the system notifies the premium user about the upcoming renewal. At the end of the period, the subscription is renewed automatically and the user is charged through the external payment system.
Primary Actors	Time
Secondary Actors	Payment system
Preconditions	1. The user has an active premium subscription. 2. The subscription has not been cancelled.
Main Flow of Events	1. Three days before the end of the current billing period, the system sends a notification to the premium user about the upcoming renewal and its price. 2. The current billing period ends. 3. The system checks that the subscription has not been cancelled. 4. INCLUDE: Process payment - the system charges the user for the next billing period. 5. The system extends the subscription by one billing period (month or year). 6. The system informs the user that the subscription has been renewed.
Alternative Flows	3a. The subscription was cancelled 1. The system does not renew the subscription. 2. The system deactivates the premium membership and the user becomes a registered user again. 3. The use case ends. 4a. The payment fails 1. The system notifies the user that the payment was not successful. 2. The system deactivates the premium membership. 3. The use case ends.
Post-conditions	1. The subscription is extended for another billing period, or 2. The premium membership is deactivated.

1.22. Report incident

ID: UC38

Use Case ID	UC38
Brief Description	If a problem occurs during a route (e.g. an accident or the vehicle cannot continue), the vehicle reports it to the system. The system immediately notifies the system operator, who takes care of the incident. The incident is stored in the vehicle's history together with its routes and route changes for troubleshooting purposes.
Primary Actors	Vehicle

Secondary Actors	System operator
Preconditions	
Main Flow of Events	
Alternative Flows	
Post-conditions	

1.23. Save payment information

ID: UC03

Use Case ID	UC03
Brief Description	Registered user can save his payment information for the next payment.
Primary Actors	
Secondary Actors	
Preconditions	
Main Flow of Events	
Alternative Flows	
Post-conditions	

1.24. Search for vehicle

ID: UC22

Use Case ID	UC22
Brief Description	User can search through available vehicles for a ride.
Primary Actors	
Secondary Actors	
Preconditions	
Main Flow of Events	

Alternative Flows	
Post-conditions	

1.25. Select car features

ID: UC10

Use Case ID	UC10
Brief Description	Premium user can select from available car features.
Primary Actors	
Secondary Actors	
Preconditions	
Main Flow of Events	
Alternative Flows	
Post-conditions	

1.26. Sending location

ID: UC23

Use Case ID	UC23
Brief Description	The vehicle sends its position to the system every second.
Primary Actors	
Secondary Actors	
Preconditions	
Main Flow of Events	
Alternative Flows	
Post-conditions	

1.27. Set availability status

ID: UC25

Use Case ID	UC25
Brief Description	Vehicle sets its availability status in the system as long as it is parked in the designated place and its battery charge is above 40%.
Primary Actors	Vehicle
Secondary Actors	-
Preconditions	1. Vehicle is parked in the designated place 2. Vehicle's battery charge is above 40%
Main Flow of Events	1. After last action (charging/order) vehicle relocates to a suitable place to park 2. Vehicle updates its position 3. Vehicle checks the battery charge level
Alternative Flows	Battery charge level below 40% 1. Vehicle is navigated to the nearest free charging station 2. Charging is disabled once the battery charge exceeds 90% 3. Vehicle is navigated to a suitable parking space
Post-conditions	

1.28. Statistics overview

ID: UC21

Use Case ID	UC21
Brief Description	Manager can view the statistics of the company.
Primary Actors	
Secondary Actors	
Preconditions	
Main Flow of Events	
Alternative Flows	
Post-conditions	

1.29. System

1.30. System operator

ID: AC11

1.31. Time

ID: AC09

1.32. Traffic information provider

ID: AC10

1.33. unnamed

Adding specifications

Q: How much is the cancellation fee?
Assumption: A fixed fee of 5 € is charged.

Q: How does the user add additional passengers to the order?
Assumption: By entering their registered e-mail addresses or username.

Q: How many offers are displayed to a premium user? Assumption: Up to 3 offers, sorted by the selection criteria.

Q: What happens if all charging stations are fully occupied?
Assumption: The vehicle is navigated to the nearest station and waits for a free slot.

Data backup is not mentioned in the specification. Agreed: the system backs up all data every day at 23:59.

1.34. Unregistered user

ID: AC02

1.35. User

ID: AC07

1.36. Vehicle

ID: AC05

1.37. View order history

ID: UC04

Use Case ID	UC04
Brief Description	Registered user can view his order history.
Primary Actors	
Secondary Actors	
Preconditions	

Main Flow of Events	
Alternative Flows	
Post-conditions	